Advancements in Detection and Management of Ductal Carcinoma in Situ

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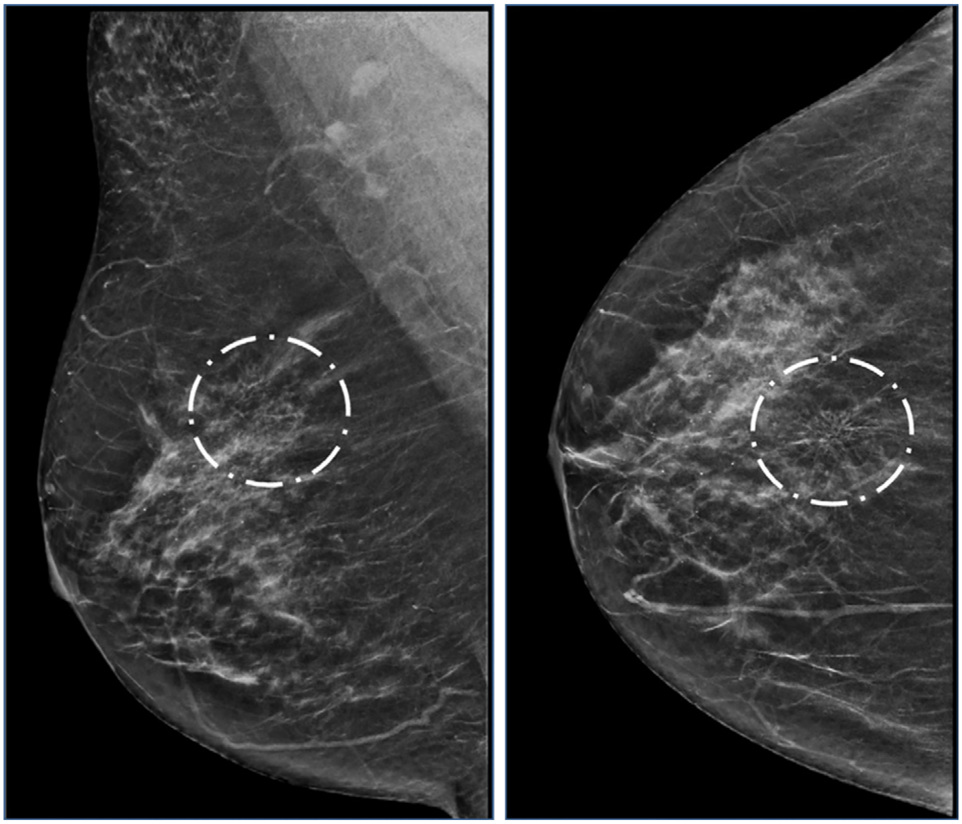
See the invited commentary by Chikarmane in this issue.



Ductal carcinoma in situ (DCIS) is a noninvasive breast cancer characterized by neoplastic epithelial cells confined to the ductal system by the basement membrane without invasion of adjacent tissue. Its progression to invasive carcinoma is not understood fully, and currently, DCIS is considered a nonobligatory precursor of invasive breast cancer. However, DCIS is challenging because it includes a heterogeneous group of lesions with varied histologic, immunohistochemical, genetic, radiologic, and clinical characteristics. This heterogeneity is reflected in its natural progression, with some lesions remaining indolent, whereas others may develop into invasive ductal carcinoma. As DCIS detection rates rise due to mammographic screening, concerns about overdiagnosis and overtreatment have emerged, which has lead to a greater focus on understanding the biologic characteristics of DCIS. Radiologists need to understand the various imaging techniques used to evaluate DCIS. These include mammography, contrast-enhanced mammography, tomosynthesis, US, and MRI. By familiarizing themselves with each modality's various strengths and limitations, radiologists can effectively assess DCIS and develop the appropriate treatment plan. Although current guidelines advise treating all cases of DCIS with surgery, radiation therapy, and hormonal therapy, ongoing trials are investigating the safety of active surveillance for women with low-risk DCIS. There is interest in improving the risk stratification of DCIS lesions, and new advanced tools, such as radiomics, artificial intelligence, and other emerging techniques, are showing positive initial results and have the potential to become valuable solutions in the future. However, further studies and development still are needed before they can be widely adopted in clinical practice.

Invited Commentary

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Abbreviations: ADH = atypical ductal dysplasia, BCS = breast-conserving surgery, CEM = contrast-enhanced mammography, DBT = digital breast tomosynthesis, DCIS = ductal carcinoma in situ, ER = estrogen receptor, FFDM = full-field digital mammography, HER2 = human epidermal growth factor receptor 2, HR = hormone receptor, IDC = invasive ductal carcinoma, PR = progesterone receptor

TEACHING POINTS

- DCIS represents a highly heterogeneous group of lesions with a spectrum of histopathologic features. Understanding this variability is important for accurate diagnosis, treatment planning, and predicting patient outcomes. The classification of DCIS based on nuclear grading highlights the complexity of diagnosing and managing this condition.
- The incidence of DCIS has increased since the introduction of breast cancer screening programs. DCIS represents a significant point of convergence in the process of breast cancer care that presents patients with both an opportunity for curative intervention and a challenge in distinguishing indolent lesions from those predisposed to progression.
- At mammography, DCIS most commonly manifests as fine pleomorphic calcifications. US, a less sensitive detection method, may show DCIS as a hypoechoic irregular mass without posterior features or as nonmass lesions. MRI most frequently demonstrates DCIS as nonmass enhancement in a segmental distribution, providing an accurate means of assessment of the extent and distribution of the disease. Each imaging technique uniquely contributes to the comprehensive evaluation of DCIS and highlights the importance of a multimodality imaging approach for accurately diagnosing and managing this condition.
- Radiologists play a critical role in the diagnosis and management of DCIS, from diagnosing and defining the extent of the disease to participating in multidisciplinary discussions on patient care.
- Ongoing trials investigate active surveillance as an alternative treatment of low-risk DCIS. This approach aims to reduce overtreatment and provide more personalized care.

Introduction

Ductal carcinoma in situ (DCIS) is the clonal proliferation of neoplastic epithelial cells confined to the duct-lobular system by the myoepithelial cell basal membrane without invading adjacent tissue (1). The progression of changes in the duct ranges from ductal hyperplasia, maintaining normal cellular morphology, to atypical ductal hyperplasia (ADH) with mild and limited atypia, progressing to DCIS with more extensive atypia but an intact basal membrane. Invasive ductal carcinoma (IDC) results from myoepithelial rupture and cellular invasion into adjacent tissue. The theoretical linear progression from ductal hyperplasia to DCIS and then IDC is shown in Figure 1. However, epidemiologic evidence suggests that DCIS is not an obligate precursor to IDC, and an alternative pathway may exist in which IDC develops directly from normal breast tissue or ADH, bypassing the DCIS stage (Fig 2) (2).

DCIS represents a highly heterogeneous group of lesions with a spectrum of histopathologic features. Understanding this variability is important for accurate diagnosis, treatment planning, and predicting patient outcomes. The classification of DCIS based on nuclear grading highlights the complexity of diagnosing and managing this condition.

DCIS lesions are highly heterogeneous in histologic, immunohistochemical, genetic, radiologic, and clinical characteristics. This diversity is shown in DCIS morphology and biologic behavior, particularly its risk of progression to invasive breast cancer (3). Despite this variability, approximately 85% of DCIS cases are asymptomatic and are detected during routine breast cancer screening programs (4). On screening mammograms, DCIS typically appears as microcalcifications. These microcalcifications can be clustered, linear, or segmental. About 15% of the time DCIS will manifest as an asymmetry, a mass, or architectural distortion. In the United States, DCIS accounts for approximately 15% of all breast cancer diagnoses, largely because of its detection through screening mammography. However, estimating the true prevalence of DCIS remains challenging because most cases are asymptomatic and are discovered only through screening (5).

The incidence of DCIS has increased since the introduction of breast cancer screening programs. DCIS represents a significant point of convergence in the process of breast cancer care that presents patients with both an opportunity for curative intervention and a challenge in distinguishing indolent lesions from those predisposed to progression.

Mammography is the most used imaging modality for detecting and characterizing DCIS. However, other modalities, such as MRI and US, also play roles in the evaluation of DCIS. Each imaging modality has its advantages and limitations in terms of sensitivity and specificity for detecting and characterizing DCIS.

The standard management of DCIS typically involves surgical excision, followed by radiation therapy to reduce the risk of local recurrence. Understanding the biology of DCIS, developments in imaging, and molecular diagnostics allows a more comprehensive approach. A multimodality method that combines radiologic, pathologic, and genetic data may lead to customized diagnosis of lesions, assessment of their aggressiveness, and guidance for treatment decisions (5).

Pathologic Features

DCIS is classified as a diverse group of lesions with varying histologic features. The basic categorization system is based on a three-tiered nuclear grading system that considers nuclear proliferation, mitotic figures, and architecture (6).

The term *nuclear grade* refers to the appearance of cancer cells, with high-grade cells exhibiting greater pleomorphism and nuclear atypia compared with low-grade cells. *Mitotic figure* indicates the proliferative activity of the tumor, whereas *architecture* describes the structural organization of cells within the mammary ducts, including comedo, solid, cribriform, papillary, flat ("clinging"), and micropapillary forms (Fig 3). These parameters are integrated into the three-tiered nuclear grading system, which classifies DCIS as low, intermediate, or high grade based on the combination of these histologic characteristics (1).

The distinction between ADH and DCIS, based on morphologic features and the extent of involvement, can be challenging for pathologists. ADH and DCIS share similar morphologic features, such as the proliferation of monomorphic epithelial cells within the ducts. However, the key difference lies in the

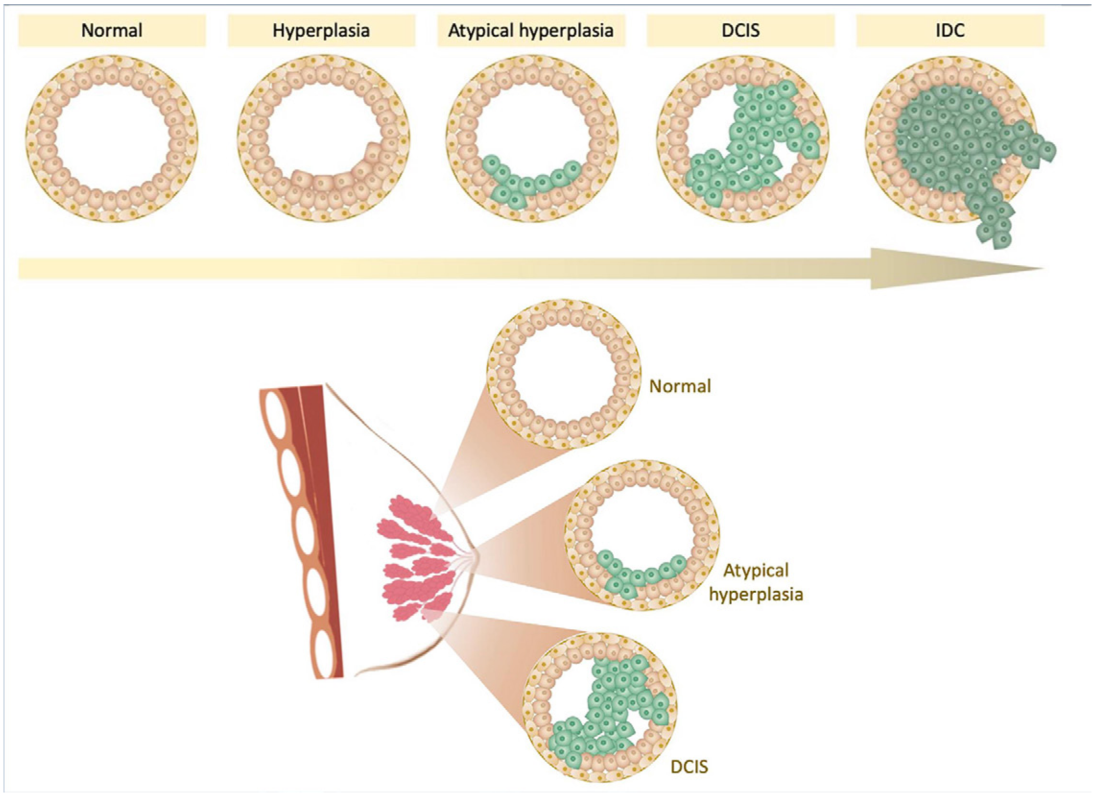


Figure 1. The normal lactiferous duct has a single layer of luminal epithelial cells, bordered by a preserved layer of myoepithelial cells and basal membrane. In ductal hyperplasia (usual), there is proliferation of epithelial cells, but these cells maintain normal morphology. ADH shows proliferation of epithelial cells as well as low-grade cytologic atypia, but the involvement of the duct is limited. In DCIS, the atypical cells of low grade that resemble ADH are more extensive, or the proliferation manifests greater cytologic atypia, and remain with an intact basal membrane. With the progression to IDC, the cytologically atypical cells increase, but there is rupture of the basal myoepithelial cell layer, which allows the invasion of carcinoma cells into the surrounding stroma.

extent of involvement. ADH typically is defined as a lesion measuring less than 2 mm or involving less than two duct spaces, whereas DCIS involves a larger area or more extensive duct spaces. The measurement of nuclear grade in DCIS is essential for determining prognosis and guiding treatment decisions. However, interobserver variability among pathologists due to the subjective assessment of morphologic features and lack of standardized grading criteria can lead to inconsistent grading and impact clinical decisions (7).

Estrogen receptor (ER) and progesterone receptor (PR) status are critical biomarkers in the diagnosis and management of DCIS. However, according to the current guidelines published by the American Society of Clinical Oncology and the College of American Pathologists, PR testing is considered optional for newly diagnosed DCIS cases without associated invasion (8). Approximately 75%–87% of DCIS cases are ER positive, which has significant implications for prognosis and treatment strategies. When ER and PR hormone receptors (HRs) are present, hormonal signals may drive the tumor cells. Adjuvant endocrine therapies, such as tamoxifen or aromatase inhibitors, can reduce the risk of recurrence by blocking the hormonal pathways that promote tumor growth. Furthermore, ER-positive DCIS tends to have a better prognosis compared with ER-negative cases, which often are more aggressive and have a higher likelihood of progressing to invasive cancer. The assessment of ER and PR status is performed routinely during the pathologic examination of DCIS (2,9).

Overexpression of human epidermal growth factor receptor 2 (HER2) is an established prognostic marker for invasive breast cancer, but the role of overexpression in DCIS remains poorly defined. HER2 testing is not performed routinely in DCIS cases. Early evidence has shown that HER2-positive DCIS cases may be associated with adverse clinicopathologic parameters and increased recurrence rates. Current clinical guidelines do not routinely recommend HER2-targeted drugs for DCIS because of the lack of conclusive evidence demonstrating a clear benefit in this noninvasive stage (2,9).

DCIS can coexist with various benign and atypical breast lesions. Intraductal papillomas may harbor areas of atypia or DCIS, whereas fibroadenomas occasionally can contain foci of DCIS. DCIS often is found alongside ADH, which shares histopathologic features with low-grade DCIS (10).

Diagnostic Underestimation and Progression

DCIS cannot be predicted to become invasive at this time. A large meta-analysis found 26% surgical upstaging (11), and a multi-institutional retrospective cohort study found 28% surgical underestimation (12). In a retrospective study of 1387

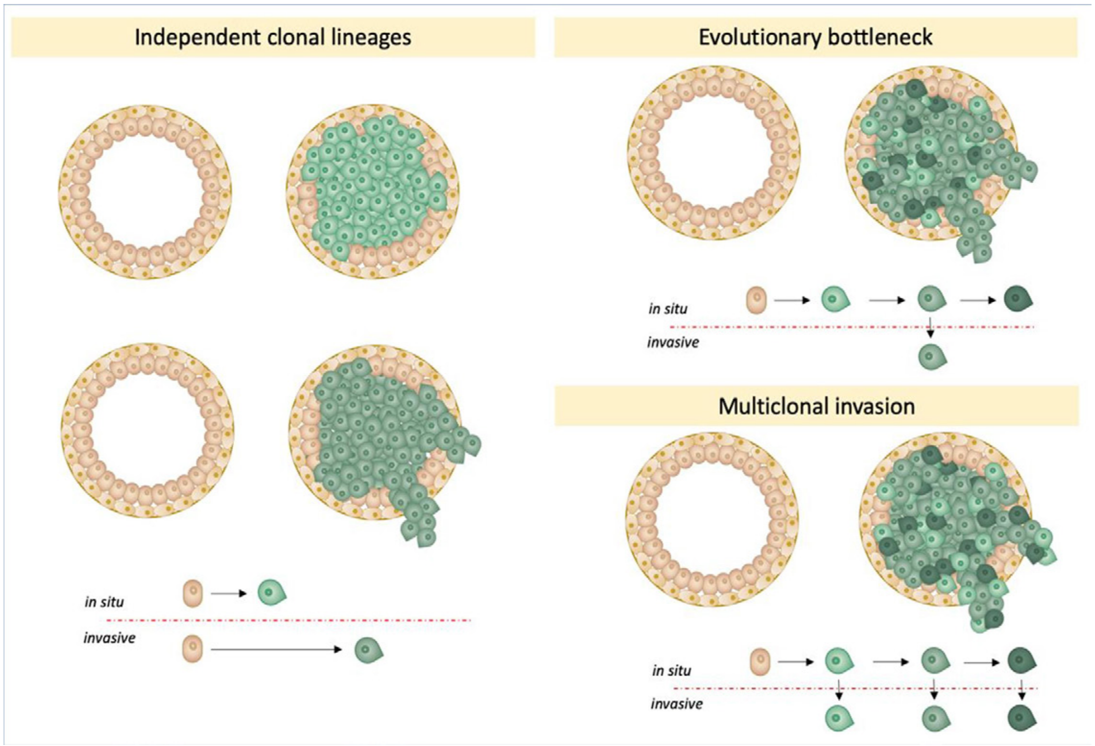


Figure 2. Evolutionary invasion models for DCIS. In the independent evolution model, DCIS and invasive cancer originate from separate normal cells without a shared clonal lineage, each undergoing distinct genetic alterations. In the evolutionary bottleneck model, multiple subclones arise within the ducts from a single cell of origin. One subclone acquires an invasive phenotype through additional mutations and breaches the basement membrane, causing an invasive tumor component. In the multiclonal invasion model, tumor cells derive from a single origin but evolve into multiple clones with genetic heterogeneity. Several clones independently acquire invasive properties, which leads to invasive disease.

women with vacuum-assisted biopsy–diagnosed DCIS, 17% were upgraded to invasion at surgery. The lower upstaging rate in this study was related to a large proportion of screening-detected DCIS with calcifications alone and the use of vacuum-assisted biopsy with larger needles (eg, 9-gauge) (13).

Risk factors associated with surgical upstaging risk include younger age, palpable findings, the presence of a mass at imaging, lesion size greater than 20 mm, intermediate or high grade, HR negativity or HER2 positivity, and the presence of comedonecrosis (4,12). Nevertheless, no clinical features are sufficiently reliable to accurately predict progression to invasive disease.

Natural History and Progression

Three invasion models are proposed during the progression from DCIS to IDC: *(a)* independent evolution, *(b)* evolutionary bottlenecks, and *(c)* multiclonal invasion. In the independent evolution model, DCIS and IDC are two different types of cells in normal breast tissue. This model contrasts with the lineage-dependent models (evolutionary bottlenecks and multiclonal evolution), which assume that a single normal breast cell gave rise to both DCIS and IDC populations. The evolutionary bottleneck model suggests that a single clone within the ducts undergoes selection during the invasion process and subsequently migrates to adjacent tissue, thereby forming the invasive tumor. Conversely, the multiclonal model suggests that invasion is facilitated by the migration of multiple clones from the duct, following the rupture of the basal membrane (2).

Mammographic Findings

Mammography is the most used imaging modality for detecting and characterizing DCIS.

Full-Field Digital Mammography

Calcifications are the predominant finding at full-field digital mammography (FFDM) and occur in more than 85% of cases without accompanying abnormalities (14,15). Other findings include masses, architectural distortions, and focal asymmetry, which may or may not be calcified.

The origin of calcifications in DCIS is attributed to the secretion of calcium within the ducts by epithelial cells and/or the calcification of necrotic debris in cases of comedonecrosis (16). DCIS typically has fine pleomorphic calcifications (~ 40%), followed by coarse heterogeneous, amorphous, fine linear or linear branching, and rare round calcifications (15) (Fig 4). These calcifications typically occur in grouped, segmental, and linear distributions (15). The fine linear or linear branching calcifications and linear or segmental distributions are more specific and visually suggest calcium deposition within the ductal system.

A recent meta-analysis (17) explored the association between morphologic descriptors and biologic characteristics of DCIS. It concluded that calcifications with fine linear, coarse

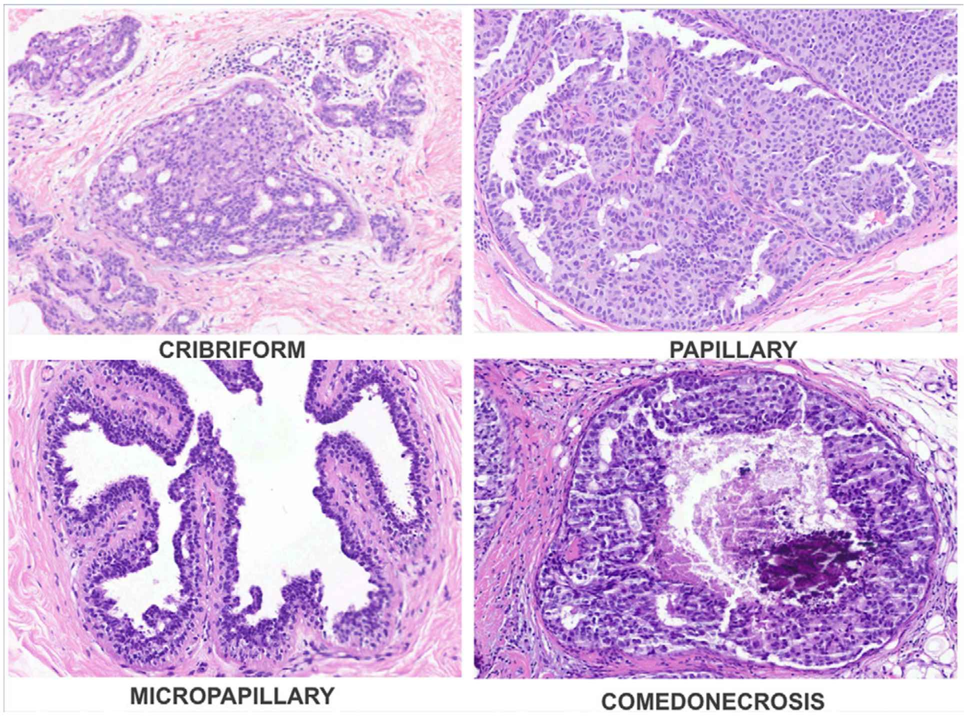


Figure 3. Clinical photomicrographs show DCIS structural organization of cells within the mammary ducts, including cribriform (hematoxylin-eosin [H-E] stain; original magnification, ×3), papillary (H-E stain; original magnification, ×4), micropapillary (H-E stain; original magnification, ×8), and comedo (H-E stain; original magnification, ×6). (Courtesy of Monica Stiepcich, MD, PhD.)

heterogeneous, and fine pleomorphic morphologies are significantly associated with high-grade DCIS and comedonecrosis compared with amorphous or round calcifications (17).

Noncalcified DCIS at FFDM typically manifests as masses (~50%), asymmetries (30%), and rarely as architectural distortion and is most commonly low grade (15). When DCIS manifests as architectural distortion, it is believed to result from sclerosing adenosis, interstitial sclerosis around the DCIS, and involvement of Cooper ligaments (2). In addition, architectural distortion frequently is caused by benign lesions, such as radial sclerosing and complex sclerosing lesions, and DCIS also can be found on the periphery of these lesions (Fig 5) (18). This understanding is essential for accurate anatomic-radiologic correlation following percutaneous biopsies.

It is noteworthy that cases of DCIS that show an overlap of noncalcified findings and calcifications demonstrate a higher likelihood of being upgraded to invasive carcinoma at surgical excision (15).

Digital Breast Tomosynthesis and Synthetic Mammography

Digital breast tomosynthesis (DBT) is a quasi–three-dimensional imaging technique that aims to improve image interpretation by reducing the effect of tissue overlap at FFDM. DBT images typically are evaluated in conjunction with images from a two-dimensional examination (eg, FFDM) obtained at the same time. They also can be evaluated in conjunction with synthetic mammographic images generated from the DBT data, with the aim of replacing the acquisition of FFDM images and reducing the radiation dose of the examination. DBT increasingly has been used in the screening and evaluation of breast cancer, which improves FFDM sensitivity and specificity, particularly for noncalcified lesions such as masses and architectural distortions (Fig 6). A study that analyzed only noncalcified DCIS demonstrated a higher sensitivity of DBT (84%) compared with that of FFDM (64%), to detect noncalcified DCIS (19). However, a large study has shown that although DBT helps identify more invasive carcinomas, there has been no improvement in the detection of DCIS (including calcified and noncalcified) (20). This could be because calcified cases still make up a larger percentage of DCIS cases detected at both DBT and FFDM, despite noncalcified findings being more clearly visualized at DBT.

Synthetic mammography has been associated with variable results in both the evaluation and detection of calcifications, as well as its performance for detecting DCIS (20–22). Although some studies have shown no differences in DCIS detection between FFDM plus DBT and synthetic mammography plus DBT, the TOMMY trial demonstrated that synthetic mammography plus DBT was inferior to both FFDM plus DBT and FFDM alone for detecting microcalcifications and DCIS in the 11–20-mm range (20–22).

Contrast-enhanced Mammography

Contrast-enhanced mammography (CEM) is an imaging modality that detects abnormal morphologic characteristics and breast cancer neovascularity, with superior performance to that of standard mammography and high sensitivity, and may be used as an alternative to MRI (23). Because CEM compensates for the lack of enhancement at mammography and the inability to show calcifications at MRI, CEM may help in the diagnosis and management of DCIS. CEM can help define

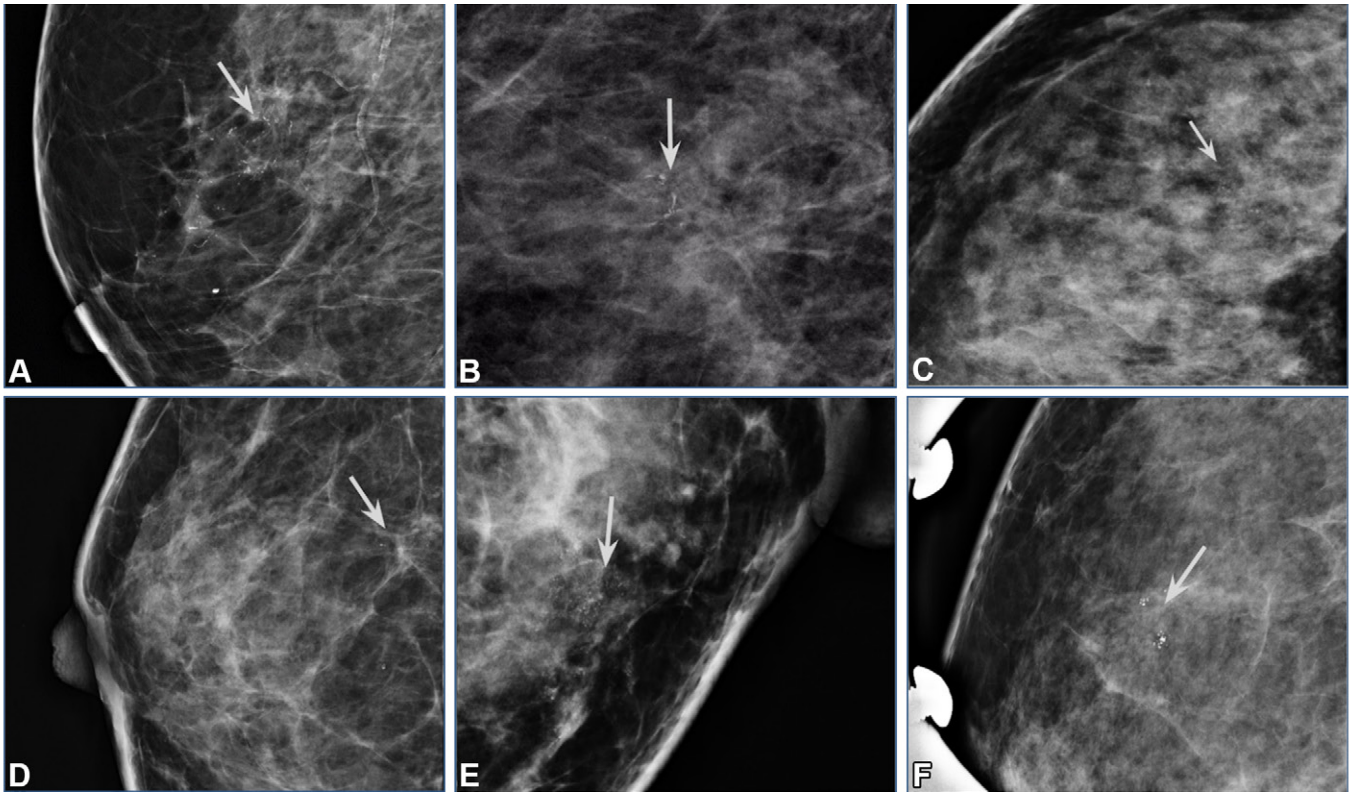


Figure 4. DCIS on magnified-view mediolateral (ML) mammograms in six patients. **(A)** Image in a 65-year-old woman shows fine-linear calcification with segmental distribution (arrow). Subsequent stereotactic vacuum-assisted biopsy results were high grade, cribriform and solid, with comedonecrosis, HR negative, HER2 negative. **(B)** Image in a 71-year-old woman shows grouped fine-pleomorphic calcification (arrow). Subsequent stereotactic vacuum-assisted biopsy results were high grade, cribriform and solid, comedonecrosis, HR positive, HER2 negative. **(C)** Image in a 36-year-old woman shows grouped amorphous and round calcifications (arrow). Subsequent stereotactic vacuum-assisted biopsy results were high grade, micropapillary and papillary, with comedonecrosis, HR positive, HER2 negative. **(D)** Image in a 56-year-old woman shows round calcification with a linear distribution (arrow). Subsequent stereotactic vacuum-assisted biopsy results were intermediate grade, micropapillary and cribriform, comedonecrosis absent, HR positive, HER2 negative. **(E)** Image in a 40-year-old woman shows grouped amorphous calcification (arrow). Subsequent stereotactic vacuum-assisted biopsy results were low grade, cribriform, comedonecrosis absent, HR positive, HER2 negative. **(F)** Image in a 40-year-old woman shows grouped round calcification (arrow). Subsequent stereotactic vacuum-assisted biopsy results were low grade, cribriform and solid, comedonecrosis absent, HR positive, HER2 positive.

accurately the extent of the disease by depicting morphologically calcified DCIS and enhancing noncalcified DCIS and invasive disease (24,25). However, not all malignant lesions demonstrate significant enhancement at CEM, including low-grade DCIS. In clinical practice, radiologists must consider the morphology and distribution of calcifications to determine the level of suspicion, even in cases in which the lesions do not enhance (24,25). Suspicious features at mammography warrant further investigation, regardless of CEM enhancement patterns.

US Findings

The ability of US to help detect DCIS is poor, with a reported sensitivity of approximately 50% (15). The typical appearance of DCIS at US image includes masses (39%–89%) and nonmass lesions (11%–58%), with or without associated calcifications (15,26,27).

Typically, masses associated with DCIS are described as solid, hypoechoic, irregular lesions with noncircumscribed margins, with absent posterior acoustic features and increased vascularity on Doppler US images (Fig 7) (15). However, cystic components may be present, and a round or oval shape with circumscribed margins also has been described (Fig 8) (15,28). DCIS can manifest as *pseudomicrocysts* at US, in which a mass with complex echotexture and microlobulated margins mimics clustered microcysts, which may be related to distension of terminal ductal lobular units (Fig 9) (29).

Although the terminology for nonmass findings at US varies across the literature, the majority of studies refer to an area of textural alteration that does not conform to a mass shape (30). These areas of textural alteration may reflect malignancy, and pure DCIS has been reported in 16% of cases (31). Nonmass findings with a segmental-linear distribution, associated ductal changes, association with calcifications, and architectural distortion are suggestive of malignancy (Figs 5, 10) (30). Other common manifestations of DCIS on US images include various ductal abnormalities, such as dilated ducts with an intraductal mass or stacked ducts (Fig 11) (32).

Although ER-positive and ER-negative DCIS have similar presentations at US, ER-negative DCIS is more likely to be sonographically visible than is ER-positive disease (15). DCIS not detected at mammography but visible at US is more likely to be of low or intermediate grade, lack comedonecrosis,

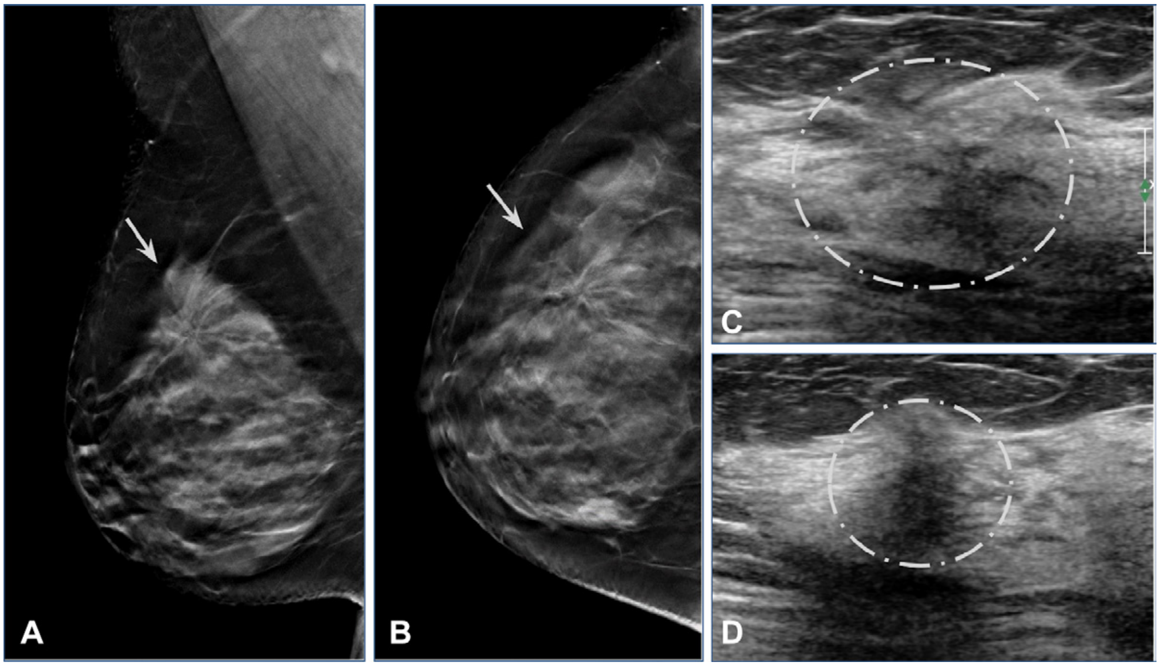


Figure 5. Asymptomatic architectural distortion detected at screening tomosynthesis. **(A, B)** Mediolateral oblique **(A)** and craniocaudal **(B)** selected tomosynthesis images show architectural distortion (arrow). **(C, D)** US images show a nonmass hypoechoic finding with focal distribution and associated architectural distortion (circle). US-guided biopsy results showed a radial sclerosing lesion (RSL) and DCIS low grade, cribriform, comedonecrosis absent, HR positive, HER2 negative. In this case, the architectural distortion is attributed primarily to the RSL rather than to the low-grade DCIS. Recognizing this distinction is crucial for accurate radiologic–pathologic correlation and for understanding the underlying pathologic condition responsible for the imaging findings.

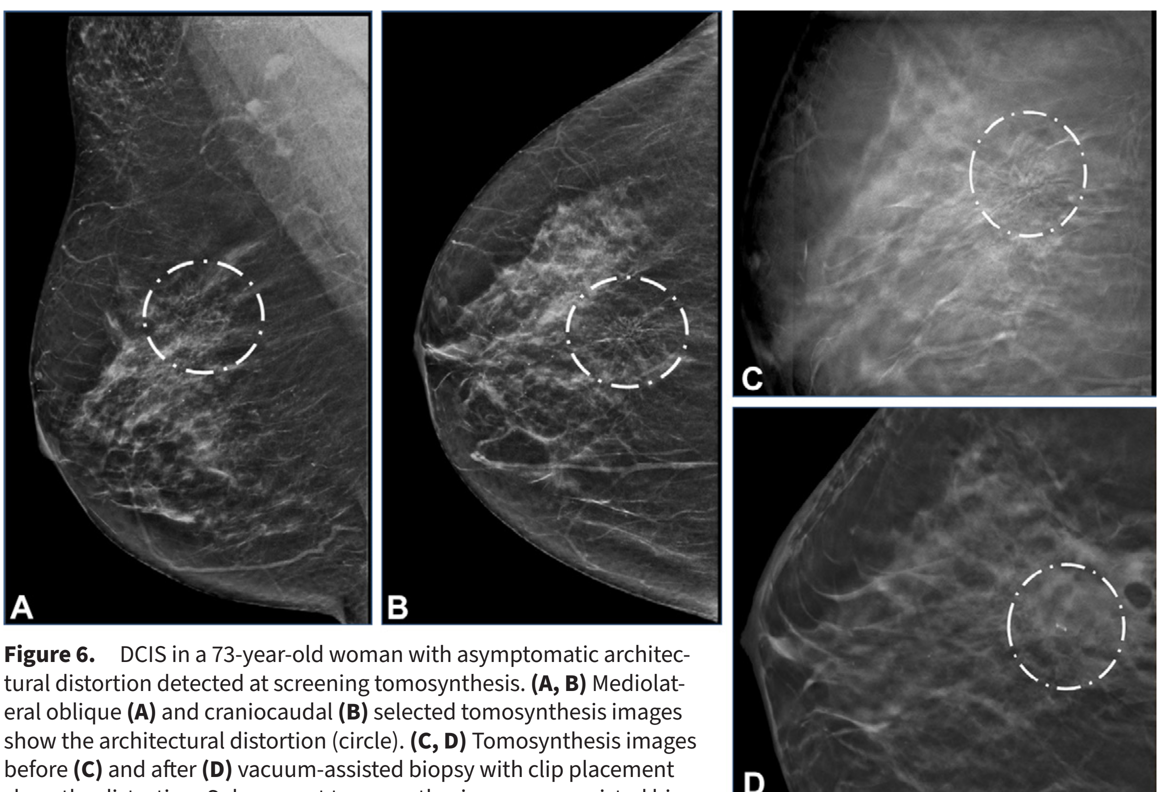


Figure 6. DCIS in a 73-year-old woman with asymptomatic architectural distortion detected at screening tomosynthesis. **(A, B)** Mediolateral oblique **(A)** and craniocaudal **(B)** selected tomosynthesis images show the architectural distortion (circle). **(C, D)** Tomosynthesis images before **(C)** and after **(D)** vacuum-assisted biopsy with clip placement show the distortion. Subsequent tomosynthesis vacuum-assisted biopsy results: DCIS low grade, cribriform, comedonecrosis absent, HR positive, HER2 negative.

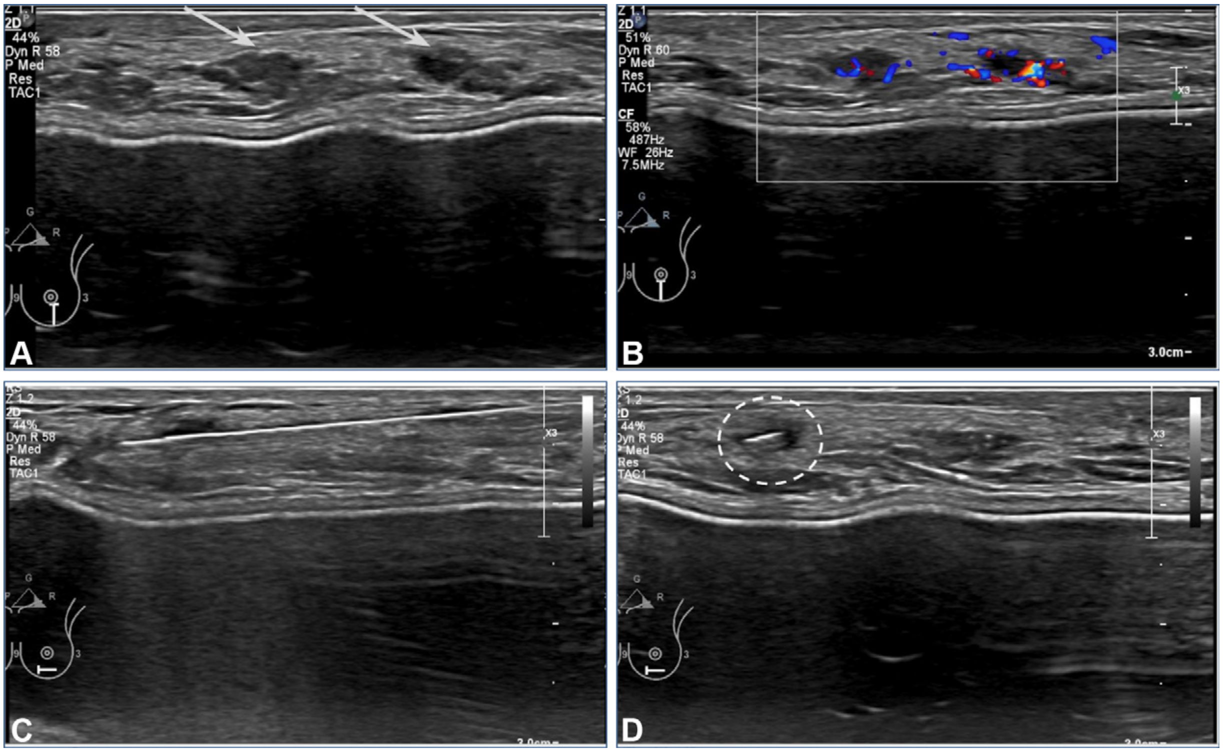


Figure 7. DCIS at breast US in a 44-year-old woman with implants and *BRCA2* genetic mutation. **(A)** US image shows two microlobulated solid masses (arrows). **(B)** Color Doppler US image shows that both masses are hypervascular. **(C, D)** US-guided core biopsy with placement of a biopsy marker (circle in **D**) was performed. Results showed high-grade DCIS, solid and papillary, comedonecrosis absent, ER positive, PR positive, HER2 positive.

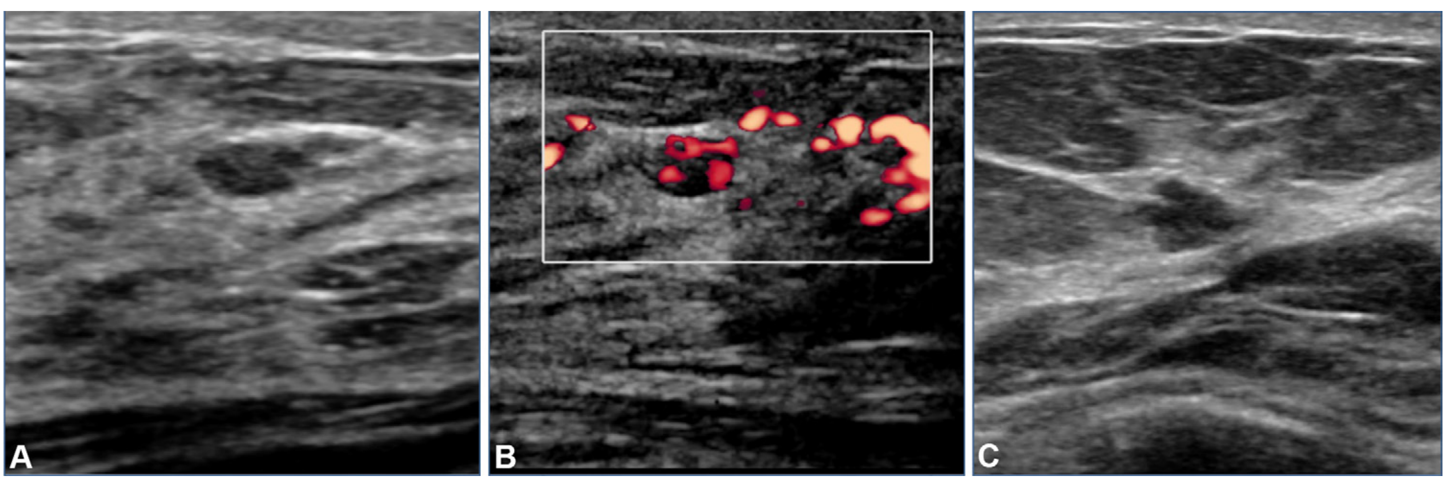


Figure 8. DCIS at US in two patients. **(A, B)** US image **(A)** in an asymptomatic 48-year-old patient with dense breasts shows a circumscribed and solid mass, and power Doppler US image **(B)** shows that the mass is hypervascular. Results of core-needle biopsy showed DCIS intermediate grade, micropapillary and cribriform, comedonecrosis absent, HR negative, HER2 negative. **(C)** US image in a different patient shows a microlobulated and solid mass. Results of core-needle biopsy showed DCIS micropapillary and papillary solid mass, low grade, comedonecrosis absent, HR negative, HER2 negative.

and lack HER2 overexpression (33). When DCIS manifests as a nonmass lesion on US images, it is more likely to be associated with a higher nuclear grade and comedo-type necrosis compared with cases in which DCIS manifests as a mass (27).

MRI Findings

The most common imaging finding of DCIS at MRI is nonmass enhancement (60%–80%), typically in a focal, linear, or segmental distribution (Figs 12–14). The morphologic distribution of DCIS is related to the growth pattern within the ductal system and the local histologic modifications determined by the neoplasm. Neoplastic cells release angiogenic factors that promote periductal vascular proliferation. The release of proteases by the neoplasm increases the permeability of the basal membrane and allows nutrients to diffuse directly into the tissue. The local increase in vascularity and vascular permeability promotes gadolinium contrast agent extravasation at the site of the neoplastic process and results in linear or segmental

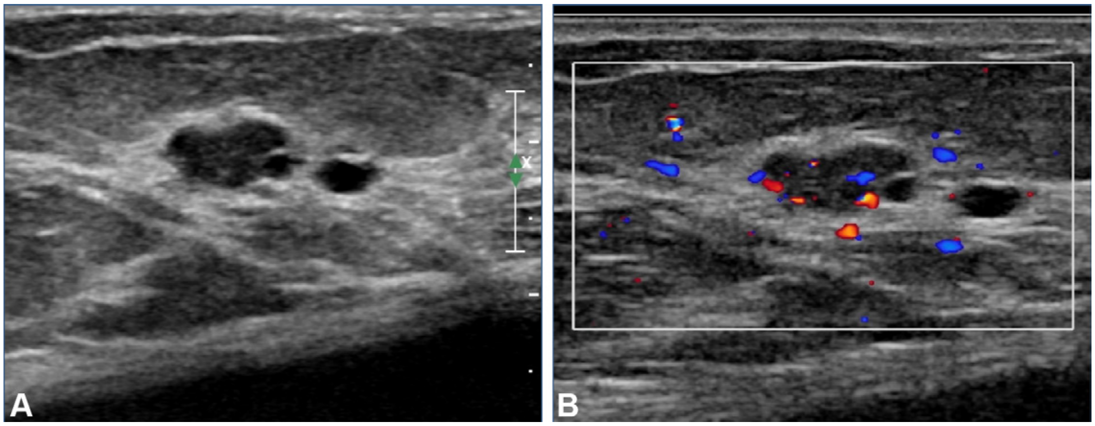


Figure 9. **(A)** US image shows a complex cystic and solid mass, mimicking microcysts. **(B)** Color Doppler US image shows internal vascularity of the mass, consistent with solid content. Core-needle biopsy showed DCIS low grade, cribriform, comedonecrosis absent, HR positive, HER2 negative.

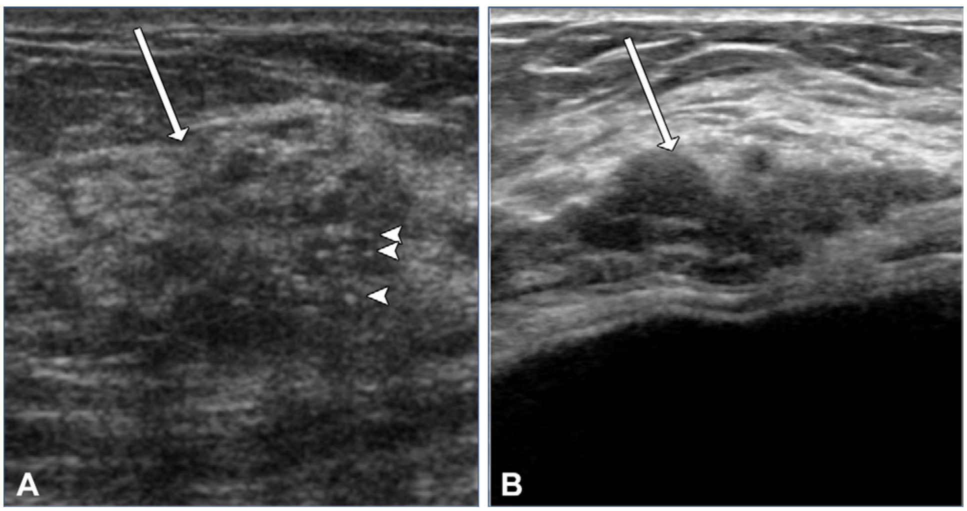


Figure 10. DCIS seen as nonmass findings at US in two patients. **(A)** US image shows a hypoechoic nonmass finding with focal distribution (arrow) and associated calcifications seen as hyperechoic foci (arrowheads). Biopsy revealed high-grade DCIS, cribriform, with comedonecrosis, HR positive, HER2 negative. **(B)** US image in a different patient shows a hypoechoic nonmass finding with segmental distribution (arrow). Biopsy revealed low-grade DCIS, micropapillary and cribriform lesion, comedonecrosis absent, HR negative, HER2 negative.

enhancement, often with a clumped or heterogeneous internal enhancement pattern (Figs 14, 15) (34). The classic example of DCIS on MR images is a segmental nonmass enhancement with a clumped internal enhancement pattern (Fig 14) (34). The clustered ring enhancement pattern is less frequent, but it is highly specific for DCIS, and its appearance likely relates to the accumulation of gadolinium contrast agent in periductal and intraductal spaces (Fig 16) (35).

DCIS can exhibit persistent, plateau, or washout kinetic enhancement patterns. Persistent enhancement, commonly associated with benign causes, can be present in 30%–52% of DCIS cases, particularly low-grade DCIS (36,37). Up to 12% of DCIS lesions do not demonstrate any detectable enhancement and typically are lower-grade lesions that manifest as microcalcifications on mammograms (38) (Fig 17). MRI been shown to contribute to overestimation of the extent of DCIS, which may be related to the enhancement of benign processes such as parenchymal background enhancement, sparse and scattered DCIS, and proliferative changes near the DCIS, such as atypia or fibrocystic changes (39). Nevertheless, MRI is the method with the best correlation with pathologic findings in the presurgical estimation of DCIS extent (40).

Imaging features at MRI associated with invasive upgrade at surgery (41–43) include masslike lesions and large lesions (42,43). Improved DCIS characterization has been shown with use of diffusion-weighted imaging and fast MRI (44). A study by Miceli et al (44) demonstrated that kinetic information from fast MRI can aid in predicting surgical upstaging to invasive cancer because these cases are associated with a shorter time to enhancement. Lee et al (45) developed a model using diffusion-weighted imaging findings and other clinicopathologic factors to predict surgical upstaging and achieved an area under the receiver operating curve of 0.76 in the validation group.

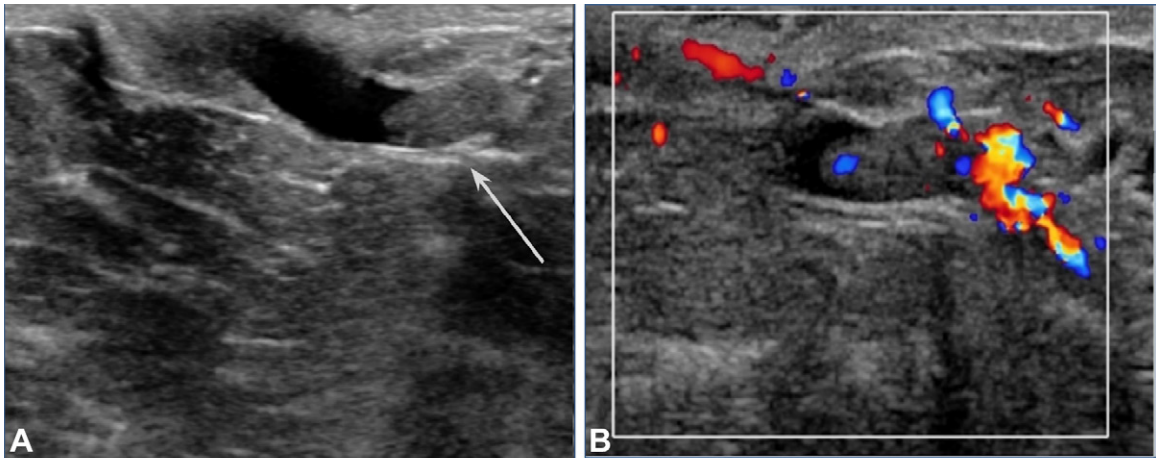


Figure 11. Intraductal mass at US. **(A)** US image shows an echogenic intraductal mass (arrow) within a dilated duct. **(B)** Color Doppler US image shows prominent vascular flow within the mass. Biopsy revealed papilloma with DCIS, low grade, cribriform, comedonecrosis absent, HR positive, HER2 negative. The intraductal nodular appearance is the aspect related to papilloma.

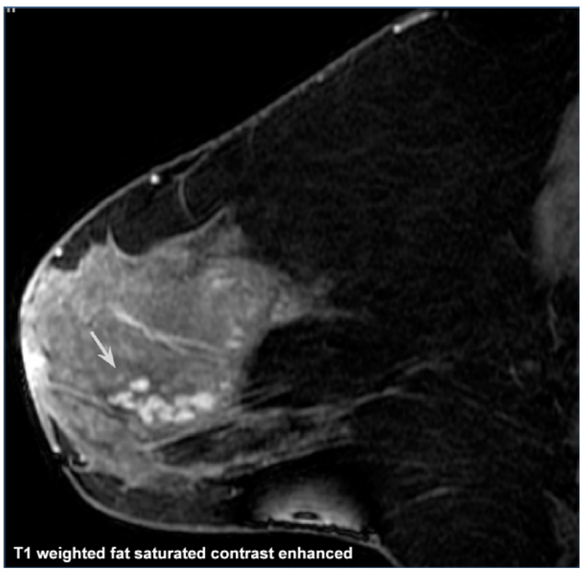


Figure 12. Sagittal contrast-enhanced T1-weighted fat-saturated MR image in an asymptomatic 46-year-old patient with dense breasts shows a focal nonmass enhancement with a clumped internal pattern (arrow). Results of MRI-guided biopsy revealed high-grade DCIS.

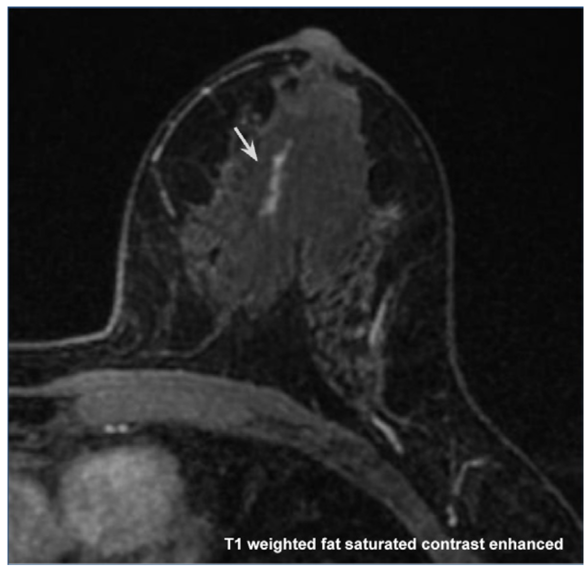


Figure 13. Axial contrast-enhanced T1-weighted fat-saturated MR image of the left breast in an asymptomatic 49-year-old patient with dense breasts shows a linear homogeneous nonmass enhancement (arrow). MRI-guided biopsy revealed intermediate-grade ductal carcinoma in situ.

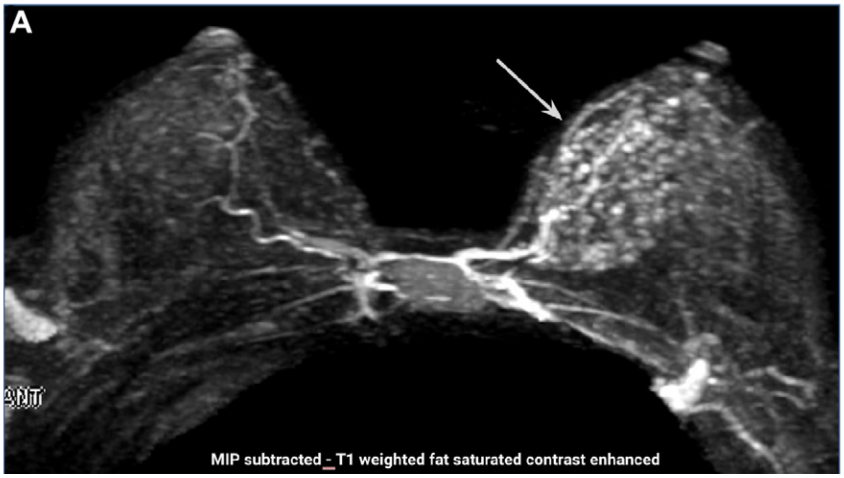
MRI features have long held promise as predictors of the biologic characteristics of DCIS. Models that combine multiple descriptive features have shown better performance in predicting the aggressiveness of DCIS than individual descriptors (46), including a multivariate model that combined contrast-to-noise ratio from diffusion-weighted images with a *b* value of 600 sec/mm^2, and maximum lesion size was shown to be the most effective feature in distinguishing high-grade DCIS from non–high-grade DCIS (area under the curve = 0.81) (46).

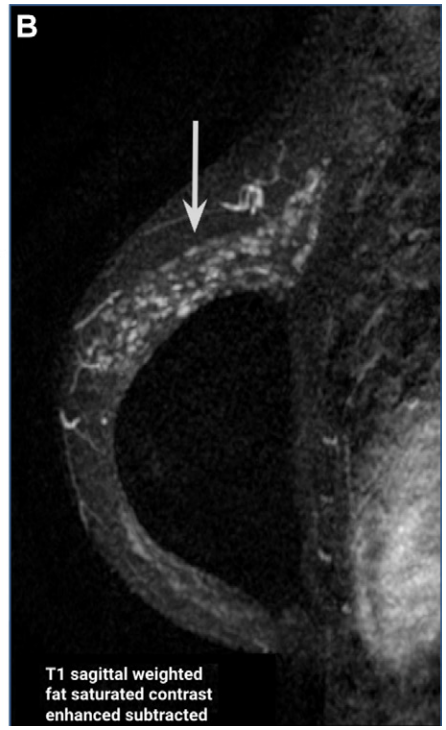
At mammography, DCIS most commonly manifests as fine pleomorphic calcifications. US, a less sensitive detection method, may show DCIS as a hypoechoic irregular mass without posterior features or as nonmass lesions. MRI most frequently demonstrates DCIS as nonmass enhancement in a segmental distribution, providing an accurate means of assessment of the extent and distribution of the disease. Each imaging technique uniquely contributes to the comprehensive evaluation of DCIS and highlights the importance of a multimodality imaging approach for accurately diagnosing and managing this condition.

Assessment of DCIS Extension

Evidence suggests that MRI is more sensitive than mammography in detection of DCIS (2,38). In a study by Kuhl et al (38), MRI was found to have a higher sensitivity for high-grade DCIS (98%) compared with that for low-grade DCIS (80%), whereas mammography's sensitivity was 52% for high-grade DCIS and 61% for low-grade DCIS. When assessing the size of

Figure 14. MRI in a 41-year-old patient with dense breasts and a family history of breast cancer. Axial maximum intensity projection **(A)** and sagittal **(B)** contrast-enhanced fat-saturated T1-weighted MR images show a segmental nonmass enhancement with a clumped internal pattern (arrow) and implants. Stereotactic-guided biopsy of calcifications in the topography revealed high-grade DCIS.





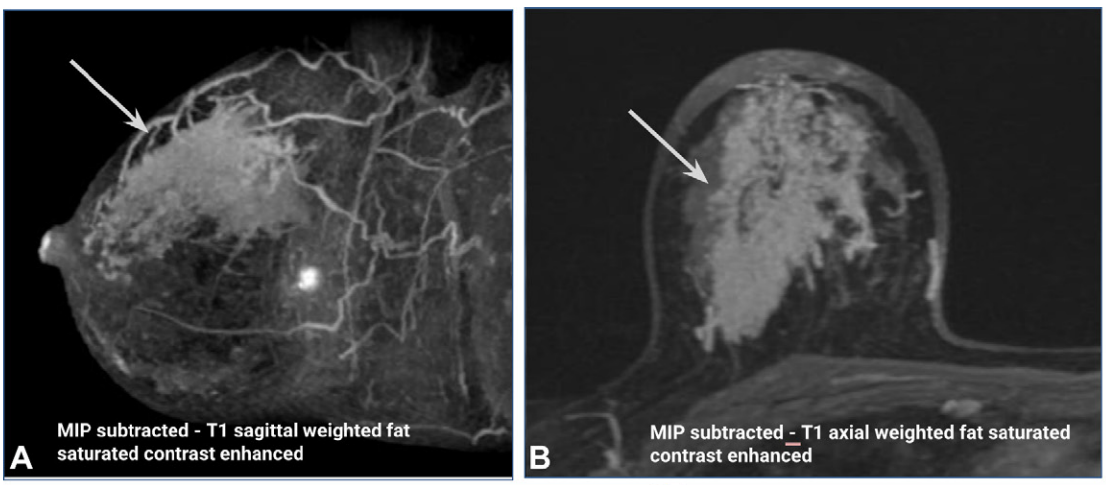
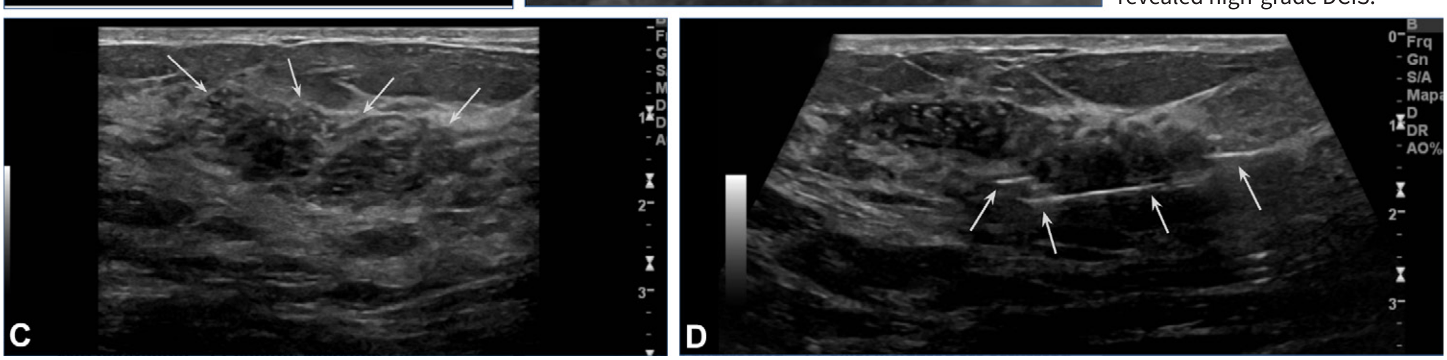


Figure 15. DCIS in a 39-year-old patient with bloody nipple discharge and dense breasts. **(A, B)** Sagittal **(A)** and axial **(B)** maximum intensity projection reconstructions of contrast-enhanced MR images show a segmental nonmass enhancement with a heterogeneous internal pattern (arrow). **(C)** US image shows a nonnodular lesion (arrows). **(D)** US-guided biopsy (arrows) was performed and revealed high-grade DCIS.



DCIS, MRI was more accurate than mammography. In a meta-analysis by Bartram et al (40), nine studies (two prospective and seven retrospective studies) met inclusion criteria to evaluate the performance of MRI in estimation of DCIS size. The majority (eight of nine) of the studies showed that MRI was more accurate than mammography when compared with findings from histopathologic analysis, with three studies demonstrating statistical significance. Six studies from the review evaluated MRI versus mammography in terms of DCIS nuclear grade, and most of them indicated that MRI was more accurate not only for high-grade DCIS but also for non–high-grade DCIS.

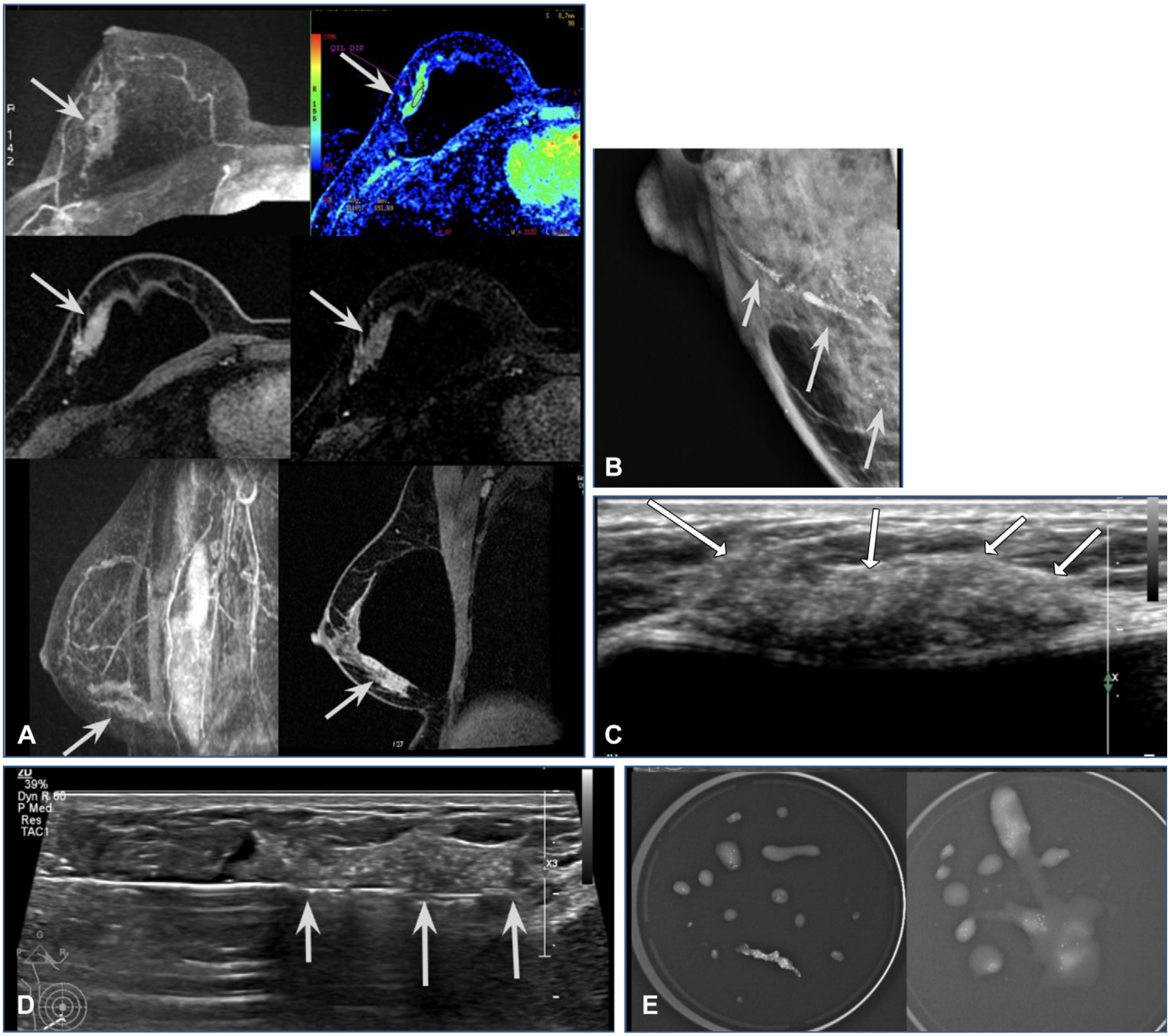
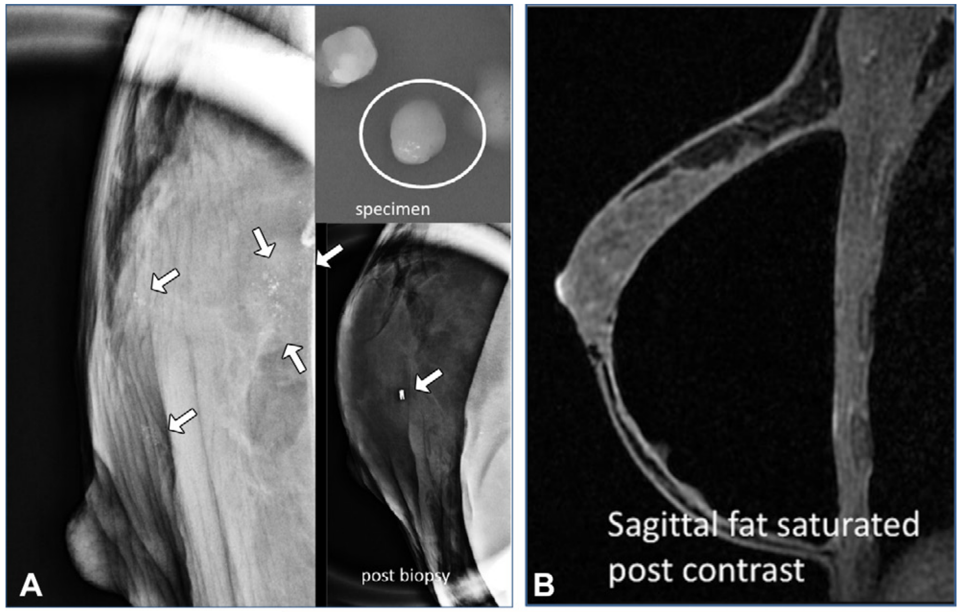


Figure 16. High-grade DCIS (grade 3) identified at MRI, mammography, and US in a 41-year-old woman with breast implants. **(A)** Axial and sagittal contrast-enhanced MR images show segmental nonmass enhancement with a clustered ring internal enhancement pattern (arrows). **(B)** Magnified-view mammogram shows pleomorphic calcifications with segmental distribution (arrows). **(C)** US image shows a hypoechoic nonmass finding with segmental distribution and associated calcifications seen as hyperechoic foci (arrows). **(D, E)** US-guided vacuum-assisted biopsy (arrows in D) was performed **(D)**, and analysis of the biopsy specimen **(E)** confirmed calcifications.

Both mammography and MRI tend to overestimate the size of DCIS when compared with pathologic analysis (47,48). Despite the tendency of MRI to contribute to overestimation, the results of a recent systematic review by Roque et al (47) showed high accuracy, with a difference of 3.85 mm when compared with the pathologic findings. Bartram et al (40) suggest that this variation in pathologic findings may be caused by the specimen's size reduction during the fixation process.

The accuracy of MRI for estimation of DCIS size is higher (Fig 18), but its preoperative use is controversial because of variable data on reoperation and mastectomy rates. Studies by Pilewskie et al (49) and Davis et al (50) showed lower but not statistically significant reoperation rates in the MRI group. Conversely, Yoon et al (51) found that preoperative MRI helped significantly reduce reoperation rates (odds ratio, 0.33; P = .03). After analyzing 10 415 DCIS cases, Keymeulen et al (52) found that preoperative MRI was linked to a higher reoperation rate (21% vs 16.7%; P < .05) compared with in the non-MRI group. In a randomized controlled trial by Balleyguier et al (53), the MRI group had a 26% lower reoperation rate (P = .13). In the observational Multicenter International Prospective Analysis (MIPA) study (54) of 1005 women with pure DCIS, the MRI group had a significantly lower reoperation rate (10% vs 22%; P = .003).



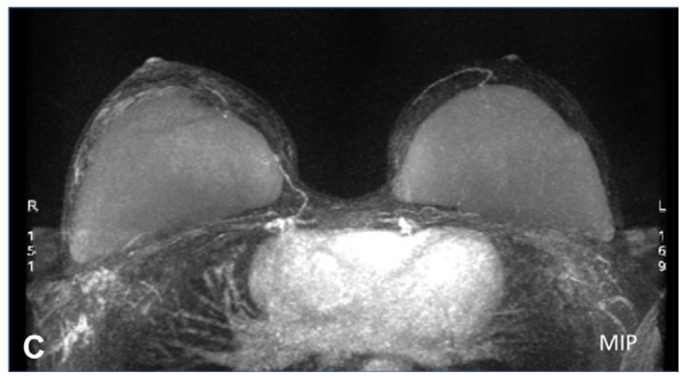


Figure 17. DCIS at MRI in a 44-year-old woman with *BRCA2* genetic mutation and breast implants. Multicentric DCIS intermediate grade (grade 2) was identified at mammography, with no enhancement at MRI. **(A)** Magnification mammogram (left) shows amorphous calcifications (arrows). Subsequent stereotactic biopsy yielded low-grade DCIS. **(B, C)** Sagittal fat-saturated postcontrast MR image **(B)** shows absence of enhancement, whereas maximum intensity projection reconstruction of contrast-enhanced MR image **(C)** shows the absence of significant enhancement.

When mastectomy rates are analyzed, the results still are variable. In a review by Bartram et al (40), eight studies that underwent statistical testing were included, with three studies showing an increased initial mastectomy rate for the MRI group, whereas five studies found no significant association. MRI and non-MRI mastectomy rates were not statistically different in the Balleyguier et al (53) randomized controlled trial (18% vs 17%; P = .93). In the MIPA study (54), the initial mastectomy rate was 20.1% for MRI and 11% for non-MRI (P = .003).

The National Comprehensive Cancer Network guidelines (55) recommend preoperative MRI for DCIS staging but note that there are insufficient data about its impact on achieving negative margins or reducing mastectomy rates.

MRI is more effective than mammography in some subgroups and may help plan treatment. These subgroups include cases where the tumor size is greater than 2 cm (56), because mammography is less accurate in these instances, as well as in younger patients (under 50 years), dense breast parenchyma (57), and cases where there are DCIS components associated with invasive carcinoma (58).

Radiologists play a critical role in the diagnosis and management of DCIS, from diagnosing and defining the extent of disease to participating in multidisciplinary discussions on patient care.

Treatment

Current best practice for treatment of DCIS is surgery and radiation therapy. Alternatives to mastectomy include breast-conserving surgery (BCS) and radiation therapy. The surgical approach depends on tumor size, DCIS grade, histopathologic subtype, patient age, and recurrence risk. For years, BCS followed by radiation therapy has been an effective treatment of DCIS (59,60).

BCS works best with small lesions, limited microcalcifications, and single lesions (61). If optimal surgical margins are not found at the time of excision, defined as 2 mm or greater, the standard practice is to perform a re-excision (62). A recent study of 197 patients by Vanni et al (61) found no association between margins under 2 mm and locoregional recurrence. The study concluded that re-excision could be avoided in patients with focally positive margins and no evidence of the disease found at postsurgical imaging.

Radiation therapy after BCS reduces recurrence risk. A mastectomy may be advised for patients with extensive or multicentric DCIS. BCS and adjuvant radiation therapy reduce local recurrence by 50% (63). After BCS, radiation therapy reduces the risk of invasive local ipsilateral breast recurrence to 5.4% from 9.5% (64).

The American College of Radiology (65) and the American Society of Clinical Oncology (66) recommend bilateral annual mammography in women with any history of breast cancer who have undergone BCS, with the first screening recommended 6–12 months after locoregional treatment and no end date. In a meta-analysis by Houssami et al (67), postexcisional mammography aided in detection of local recurrences with 80.5% sensitivity. Moreover, postexcisional mammography allows evaluation of the response of breast tissue to adjuvant radiation therapy.

Sentinel lymph node biopsy is not indicated for patients with noninvasive disease undergoing BCS. However, for patients requiring mastectomy, sentinel lymph node biopsy is advised (68). Limited data exist on the incidence of upgrade

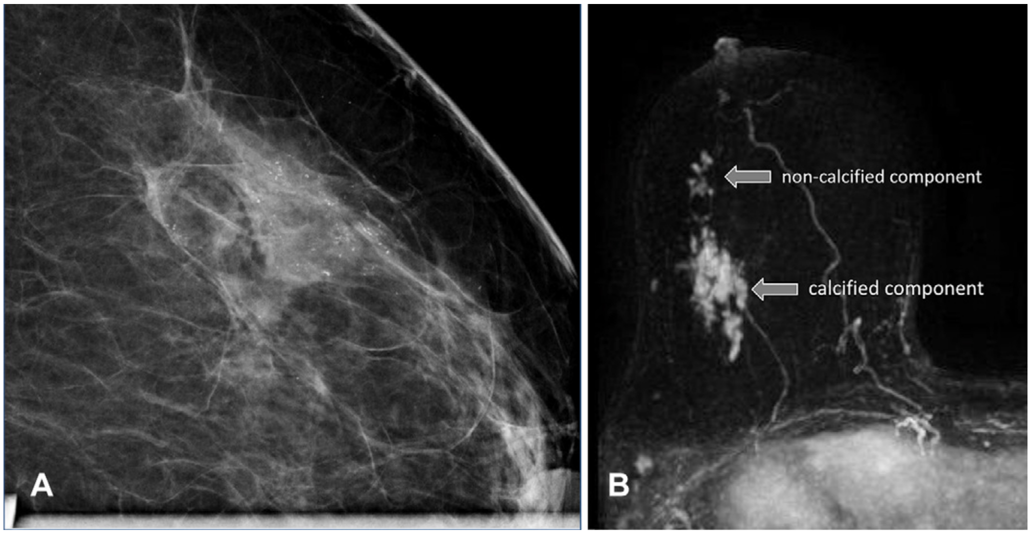


Figure 18. Comparison between mammographic and MRI findings and extension. **(A)** Magnified-view mammogram shows thin pleomorphic calcifications with segmental distribution. **(B)** Maximum intensity projection reconstruction of contrast-enhanced MR image shows a nonmass enhancement with segmental distribution and a clumped internal pattern. The enhancement includes a component corresponding to the calcifications, as well as a noncalcified component discernible solely at MRI.

to invasive breast cancer and axillary lymph node metastasis in patients with DCIS undergoing mastectomy. A recent study of 385 patients found that axillary lymph node metastasis was rare, with multivariable analysis identifying age 50 years or less as an independent associated factor. These results suggested that patients over 50 years of age and/or without suspicious axillary lymph nodes at radiologic evaluation could skip sentinel lymph node biopsy (68).

Endocrine therapy is recommended for ER-positive DCIS because it reduces the risk of recurrent breast cancer but does not significantly impact overall survival (69). Tamoxifen therapy reduces IDC recurrence in patients with ER-positive DCIS. In combination with radiation therapy, tamoxifen did not improve outcomes (70). For postmenopausal patients with HR-positive DCIS, anastrozole has comparable efficacy and side effects to those of tamoxifen (71). Anastrozole may be better for those with tamoxifen contraindications, such as endometrial cancer or thromboembolic events (71).

Future DCIS management will focus on distinguishing progressive from nonprogressive forms, which can lead to overtreatment with limited benefit. Multiomics, artificial intelligence, and other emerging technologies can create a risk stratification system for personalized treatment and reduce overtreatment.

Active Surveillance Studies

The natural history of DCIS is not well defined because most patients diagnosed with the disease are traditionally treated. Patients with DCIS are treated to prevent IDC and metastatic disease. To reduce breast cancer mortality, current guidelines recommend surgery, radiation, and hormonal therapy for all cases of DCIS. However, screen-detected DCIS may have an indolent course, with no or slow growth during the patient's lifetime (5). Long-term epidemiologic studies have shown that population screening programs did not significantly reduce advanced invasive disease after DCIS detection. A study from the National Health Service in the United Kingdom showed that for every three cases of screen-detected DCIS treated, there was just one fewer invasive carcinoma in the next 3 years (72). Based on these findings, active surveillance may be a safe alternative to surgery for women with low disease-progression risk (5). Four prospective randomized clinical trials (Table) are ongoing to evaluate whether active surveillance is safe in women with low-risk DCIS. For all studies, enrolled women were randomly assigned to either surgery (plus radiation therapy and endocrine therapy) or nonsurgical active monitoring. The primary endpoint is to evaluate invasive cancer progression and quality of life. These studies aim to help low-risk DCIS patients choose between standard therapies and active monitoring.

Ongoing trials investigate active surveillance as an alternative treatment of low-risk DCIS. This approach aims to reduce overtreatment and provide more personalized care (Table).

The COMET (Comparison of Operative versus Monitoring and Endocrine Therapy) trial for low-risk DCIS opened in the United States in June 2017. Eligibility criteria included 40 years of age or older, pathologic confirmation of grade I or II DCIS by two pathologists, mammographic calcifications without a mass at physical examination or imaging, ER, and/or PR greater than or equal to 10%. This study had the option of endocrine therapy in both arms. The primary goal is to assess if the 2-year ipsilateral invasive breast cancer rate for active monitoring is noninferior to that for surgery. Surveillance consists of a clinical examination every 6 months for a minimum of 5 years and every 12 months thereafter for both groups. Those in the active monitoring group were followed with ipsilateral mammography every 6 months and contralateral mammography every 12 months. According to the COMET trial, radiographic criteria for potential DCIS progression and indication for biopsy included

Table: Summary of Active Surveillance Trials for DCIS

Component	COMET	LORIS	LORD	LORETTA
Trial design	RCT	RCT	RCT or patients' choice	Single arm
Year initiated	2017	2014 and 2016	2017	2017
Enrolling locations	100 sites in United States	United Kingdom	52 in European Union	Japan
Control arm	Standard local control	Standard local control	Standard local control	...
Study arm	Active surveillance and choice of endocrine therapy	Active surveillance	Active surveillance	Active surveillance and mandatory endocrine therapy
No. of women planned	1200	932	1240	340
Age (y)	≥40	≥46	≥45	≥40
Size	Any size	NS	Any size	≤2.5 cm
Nuclear grade	1 or 2	1 or 2	1	1 or 2
Primary outcome	Ipsilateral invasive cancer at 2-y follow-up	Ipsilateral invasive breast cancer–free survival at 5 y	Ipsilateral invasive breast cancer–free survival at 10 y	5-y cumulative incidence of invasive ipsilateral breast cancer
Imaging follow-up	MMG every 12 mo (control arm) MMG every 6 mo (study arm)	MMG every 12 mo	MMG every 12 mo	MMG or US every 6–12 mo

Sources.—References 73–76.
Note.—MMG = mammogram, NS = not specified, RCT = randomized controlled trial.

the following: an increase in the extent of calcifications greater than 5 mm in at least one dimension compared with the most recent prior mammogram, a new mass or architectural distortion on the surveillance mammogram, or new suspicious findings on other radiologic studies (eg, US, MRI). Women were stratified by age (<55 years, 55–65 years, >65 years) as well as by the maximum diameter of microcalcifications (<2 cm, 2–5 cm, >5 cm) (73).

The LORIS (LOw Risk dcIS) trial is a multicenter study that opened to recruitment in July 2014. Eligibility criteria included the following: women 46 years of age or older, screen-detected low- and intermediate-grade DCIS of any size and confirmed after vacuum-assisted core-needle biopsy for calcifications alone, and central pathologic review. Active monitoring should be done with annual mammography for at least 10 years. Patients were randomized to standard surgery or active monitoring, and no antiestrogen treatment was offered (74).

The LORD (LOw Risk DCIS) trial included women aged 45 years or older with screen-detected low-grade DCIS (grade 1) that was diagnosed on vacuum-assisted biopsy for calcifications only. The active monitoring included annual mammography for 10 years (75). However, this randomized trial changed its design to patient preference in July 2020 because most women decided that the treatment of low-risk DCIS should be a patient's choice instead of randomization. A recent publication with 377 women enrolled in this trial showed that 76% chose active surveillance for low-grade DCIS instead of surgery.

The COMET trial closed patient accrual in December 2022, and the LORIS trial closed in April 2020, both with a lower recruitment target. Data will be reported for up to 10 years from the randomization and follow-up of all recruited women. So far, there are no published data regarding LORIS and COMET.

The LORETTA trial is a single-arm trial of endocrine therapy alone for ER-positive low-risk DCIS that is ongoing in Japan (JCOG 1505). Inclusion criteria were women with DCIS nuclear grade 1 or 2, no comedonecrosis, ER positive, HER2 negative, who will receive endocrine therapy with 20 mg of tamoxifen daily for 5 years. The primary outcome is the 5-year cumulative incidence of invasive ipsilateral breast tumor. In this trial, findings other than calcification at mammography also are eligible, such as nonnodular findings at breast US. Women will be followed up for 10 years with mammography and US every 6 months for the 1st year and annually thereafter. There are no preliminary results for this study (76).

To implement an active surveillance program, it is crucial to select eligible patients carefully. Women at low risk of invasive breast cancer should be chosen. Women who are younger, are Black, have large lesions, have DCIS with a high nuclear grade, are HR negative, or have HER2 positivity should receive the standard of care treatment and not undergo active surveillance. Pilewskie et al (77) studied 296 women who met LORIS eligibility criteria and underwent surgical excision from 2009 to 2019 in a large American cancer center to determine the incidence and characteristics of synchronous invasive carcinoma found in this subset of patients (ie, women >46 years of age with screen-detected calcifications and non–high-grade DCIS diagnosed by core biopsy). The overall upgrade rate was 20%, even among this highly selected patient population that met LORIS eligibility. Upgrade to invasive carcinoma at surgical excision was much higher among women with intermediate-grade DCIS at core biopsy (23%) compared with among those with low-grade DCIS (7%). Also, patients undergoing mastectomy were more likely to have synchronous invasive disease, likely related to extent of the calcifications at initial presentation. The authors suggest that the LORIS criteria do not adequately define the low-risk DCIS population and conclude that until additional risk stratification is available to identify a population at lower risk for synchronous invasive carcinoma, surgical excision should remain as initial management of non–high-grade DCIS diagnosed by core biopsy (77). Patients who choose active surveillance must undergo close screening to detect progression to invasive disease early. Treatment and outcomes do not differ substantially from those at the time of DCIS diagnosis (5).

Common exclusion criteria are the presence of invasive or microinvasive cancer, a prior history of breast cancer, and specific pathologic characteristics that do not meet the study's inclusion criteria. Patients are removed from studies if invasive cancer is detected during follow-up, if new medical conditions arise that could interfere with the study's safety or outcomes, or if the patient withdraws consent.

Results from the COMET and LORIS trials should provide additional data about the safety of observation alone for a select cohort of women with low-grade DCIS diagnosed by core needle biopsy, such as only low-grade DCIS with a small extension of mammographic calcifications diagnosed after vacuum-assisted biopsy.

Prognosis

DCIS in itself is not a life-threatening disease. However, it can progress to IDC and gain the ability to metastasize (78). When left untreated, between 25% and 60% of DCIS cases may progress to IDC over time (79). Data from a cohort study of 144 524 women treated for DCIS (78) showed a 3.36-fold increased risk of mortality for breast cancer compared with that in women without DCIS, and the elevated risk of death persisted more than 15 years after diagnosis. Survival rates at 10 and 20 years were 98.9% and 96.7%, respectively. With recurrence, 50% of patients present with invasive disease. Overall, deaths from breast cancer in women treated for DCIS occurred because of recurrence as IDC or IDC present but not detected and treated at the time of diagnosis (78).

De-escalation of Therapy

One strategy to reduce overtreatment of DCIS involves the de-escalation of post-BCS radiation therapy, with decreases in the duration of therapy, the extent of the breast to be radiated, and even elimination of radiation therapy regimens. A study of 186 patients with DCIS who had BCS alone showed a 10-year disease-free survival rate of 94% for those with low-risk DCIS and 83% for those with intermediate- or high-risk DCIS (80). A recent study randomized 636 patients with low-risk DCIS to either radiotherapy or observation after surgery and showed that with a median follow-up of 7 years, local recurrence was lower after radiation therapy than after observation (0.9% vs 6.7%) (81). Finally, patients with newly diagnosed DCIS have a 20% risk of upgrading to invasive carcinoma after surgical resection, which requires caution in de-escalating DCIS treatment (82).

Artificial Intelligence and DCIS

Deep learning models have shown significant potential for diagnosing breast cancer with data from imaging modalities such as mammography, US, and MRI. These deep learning–based radiomic techniques can enhance the diagnosis, prognosis, and treatment monitoring of breast cancer (83). Deep learning is a subset of machine learning and artificial intelligence that uses deep neural networks to analyze complex medical data, particularly in breast cancer diagnostics.

In a retrospective study involving 420 women, an artificial intelligence model was used to distinguish between minimally invasive breast cancer and DCIS by integrating data from mammography, US, and histopathologic findings. The model achieved an area under the curve of 0.93. This high value indicates that the artificial intelligence model has a 93% probability of accurately distinguishing between DCIS and minimally invasive breast cancer and demonstrates its effectiveness in improving diagnostic accuracy (84).

Other studies have explored the development of convolutional neural networks to distinguish between low-grade and high-grade DCIS, as well as their correlation with IDC. These studies reported an area under the curve of 0.76, which is higher than for previous models that did not use convolutional neural networks. This higher area under the curve indicates improved accuracy in identifying high-risk DCIS cases (85). These studies can assist in selecting low-risk patients for clinical trials, refining decision making, and therapeutic planning. Despite these promising results, the adoption of artificial intelligence techniques in clinical practice has been slow (83).

Conclusion

DCIS represents a complex and heterogeneous spectrum of lesions, distinguished by diverse histologic, genetic, and immunohistochemical profiles. Its radiologic presentation mirrors this diversity, ranging from mammographic calcifications suggesting calcium deposition within the ductal system to nonmass lesions visible at MRI.

In contemporary practice, merely recognizing these varied imaging features is insufficient. A comprehensive understanding of the advantages and limitations of each imaging modality in the detection and evaluation of DCIS is critical. With this knowledge, radiologists can effectively contribute to accurate identification and optimal therapeutic planning, such as ensuring that the imaging-based assessment of disease extent is adequate or identifying cases with potential for surgical upgrade.

Moreover, the biologic behavior of DCIS is still not fully understood, and we are eager to differentiate between indolent lesions and those with a high risk of progressing to IDC. Active surveillance trials are anticipated to yield crucial insights into de-escalating therapy in select cases, potentially redefining treatment paradigms. Similarly, the search for biologic, clinical, and imaging markers to better stratify the aggressiveness of each case of diagnosed DCIS has shown promising results, particularly through the adoption of artificial intelligence and advanced imaging techniques. These developments are anticipated to support the implementation of personalized medicine that reduces overtreatment and more effectively manages high-risk lesions.

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